

Software Test Plan (STP)

Railway Ticket Booking System with Cancellation Feature

Document ID	STP-RTBS-2026-V1	Date	15-09-2026
Version	1.0 (Final)	Status	Approved for Execution
Authors	Quality Assurance Team	Target System	Railway Ticket Booking System (RTBS) v1.0

1. Introduction

1.1 Purpose: This document defines the master software test plan for the **Railway Ticket Booking System (RTBS) v1.0**. It details the testing strategy, operational scope, resource allocation, environment configs, risk management, and execution schedule for validating booking and cancellation functionalities.

1.2 Scope: Verification covers authentication, train search, real-time inventory locking, payment processing, PNR generation, full and partial ticket cancellation, waitlist automatic promotion, and refund calculations. Hardware firmware and third-party bank internal core engines are excluded.

1.3 References: RTBS SRS v1.0, RTBS Architecture Specification v1.0, PCI-DSS v4.0 Standard.

1.4 Definitions: PNR (Passenger Name Record), RTM (Requirements Traceability Matrix), UAT (User Acceptance Testing), STP (Software Test Plan).

2. Test Items

- User Authentication & Session Subsystem
- Train Schedule & Seat Inventory Module
- Reservation & PNR Generation Engine
- Cancellation Processing Module (Full & Partial)
- Refund Strategy & Penalty Calculation Engine
- Payment Gateway Integration Adapter
- Waitlist Management & Auto-Promotion Engine

3. Features to be Tested

Requirement ID	Feature Description	Target Verification Objective
RTBS-F-001	User Login & Session Security	Verify JWT token issuance, password security, and RBAC permissions.
RTBS-F-010	Train Search & Inventory Query	Verify schedule lookup speed and seat counts across classes.

Requirement ID	Feature Description	Target Verification Objective
RTBS-F-020	Ticket Reservation & Payment	Verify seat isolation, lock expiration (10 min), and transaction confirmation.
RTBS-F-030	Full Ticket Cancellation	Verify complete ticket cancellation, seat release, and refund initiation.
RTBS-F-031	Partial Ticket Cancellation	Verify cancelling selected passengers in a multi-passenger PNR.
RTBS-F-040	Dynamic Refund Rule Engine	Verify penalty tiers (>48 hrs, 12-48 hrs, <12 hrs prior to departure).
RTBS-F-050	Waitlist Auto-Promotion	Verify waitlisted passengers are auto-promoted when a confirmed seat is cancelled.
RTBS-NF-001	Performance & Load	Verify ≤ 2 sec response time under 5,000 concurrent transactions.

4. Features Not to be Tested

- Core banking payment gateway internal ledger processing (Vendor Responsibility).
- Physical railway station kiosk / hardware printing mechanisms.
- Third-party SMS/Email gateway network infrastructure delivery guarantees.

5. Test Approach / Strategy

Testing Levels:

- **Unit Testing:** JUnit 5 and Mockito for business rules and refund penalty calculations.
- **Integration Testing:** Spring Boot Test containers validating PostgreSQL queries and Redis locks.
- **System Testing:** End-to-end user workflows from search through booking, cancellation, and refund.
- **User Acceptance Testing (UAT):** Final validation by business stakeholders and passenger focus groups.

Testing Types: Functional, Regression, Performance/Load (JMeter), Security/Penetration Testing, Usability.

Entry Criteria: Code freeze achieved, build deployed to QA staging, test data seeded, unit test pass rate $\geq 95\%$.

Exit Criteria: 100% test case execution, zero Critical/Blocker bugs open, 100% traceability for functional requirements.

5.1 Security & Compliance Validation

- Sanitise log outputs to verify zero storage of sensitive PCI-DSS payment data.
- Validate protection against OWASP Top 10 vulnerabilities (SQLi, XSS, CSRF, Broken Auth).
- Perform authorization fuzzing to ensure users cannot cancel tickets belonging to other PNRs.

6. Test Environment

Component	Specification / Configuration
Application Servers	Linux Ubuntu 22.04 LTS, 4 vCPU, 16 GB RAM (Containerized Docker Instances)
Database / Cache	PostgreSQL 16 Multi-AZ, Redis 7 Cluster (3 Nodes)
Automation Tools	Selenium WebDriver (UI Automation), Postman / RestAssured (API Testing)
Performance Tools	Apache JMeter (Load Simulation), Grafana / Prometheus (Resource Monitoring)
Defect Management	Jira Software with Xray Test Management Integration

7. Test Schedule

Milestone / Phase	Start Date	End Date	Status
Test Case Design & Review	16-09-2026	20-09-2026	PLANNED
Environment Setup & Data Seeding	21-09-2026	22-09-2026	PLANNED
Cycle 1 Test Execution (Core Flows)	23-09-2026	30-09-2026	PLANNED
Cycle 2 Test Execution (Cancellation & Refund)	01-10-2026	07-10-2026	PLANNED
Performance & Security Hardening	08-10-2026	12-10-2026	PLANNED
User Acceptance Testing (UAT)	13-10-2026	18-10-2026	PLANNED

8. Test Deliverables

- Master Software Test Plan (STP) Document
- Comprehensive Manual & Automated Test Suite Scripts
- Requirements Traceability Matrix (RTM)
- Defect Logs & Triage Summary Reports
- Final Test Sign-off & System Quality Assessment Report

9. Roles and Responsibilities

Role	Name	Key Responsibilities
QA Lead	<Name>	Overall test plan ownership, schedule management, environment coordination.
Test Automation Engineer	<Name>	Develop REST API and Selenium UI scripts, execute load tests via JMeter.

Role	Name	Key Responsibilities
Security Test Engineer	<Name>	Perform penetration testing, OWASP validation, and PCI-DSS compliance checks.
Dev Lead	<Name>	Bug triage support, fixing reported defects, deploying staging builds.
Product Owner	<Name>	UAT acceptance criteria approval and final release sign-off.

10. Risks and Mitigation Strategy

Risk Event	Potential Impact	Mitigation Plan
Payment Gateway Sandbox instability	Blocks automated refund execution testing	Implement mock Payment Gateway stubs to emulate success/failure scenarios.
Late build delivery containing cancellation logic	Shortens QA execution timeline	Prioritize smoke and high-priority cancellation test cases; run automated suites overnight.
Race condition bugs in multi-passenger cancellation	Corrupts seat inventory counts	Utilize JMeter multi-threaded tests simulating simultaneous cancellations.

11. Assumptions & Dependencies

- Staging test environment mirrors production database schema and Redis cluster setup.
- Payment Gateway sandbox APIs support simulated mock refunds and chargebacks.
- All test data (train schedules, routes, user profiles) will be pre-populated before execution.

12. Suspension & Resumption Criteria

Suspension Criteria: Execution will be suspended if a Blocker defect prevents access to primary booking/cancellation workflows, or if the test environment is down for over 4 hours.

Resumption Criteria: Testing resumes once the blocking issue is resolved, re-verified via a successful smoke test, and deployed to staging.

13. Test Case Management & Traceability Matrix (RTM Sample)

Req ID	Test Case ID	Test Case Summary	Expected Result
RTBS-F-030	TC-CAN-01	Full Cancellation > 48 hours prior to train departure	Flat \$20 fee deducted; balance credited. Seat status released.
RTBS-F-030	TC-CAN-02	Full Cancellation < 12 hours prior to train departure	50% fare penalty applied; 50% credited to user.
RTBS-F-031	TC-CAN-03	Partial Cancellation (Cancel 1 out of 3 passengers on PNR)	1 seat released; original PNR updated for remaining 2 passengers.
RTBS-F-050	TC-WAIT-01	Auto-promotion of Waitlist passenger upon Cancellation	Waitlist Passenger #1 promoted to Confirmed seat automatically.

14. Test Metrics & Reporting

- **Tracked Metrics:** Test Execution Pass Rate (%), Defect Density per KLOC, Open Defects by Severity, Requirements Coverage (%).
- **Reporting Cycle:** Daily Execution Status Summary sent via Email/Slack; Final Test Completion Report produced post-UAT.

15. Document Approvals

Role	Name / Title	Signature & Date
QA Lead	<Name>	Approved (15-09-2026)
Development Lead	<Name>	Approved (15-09-2026)
Product Owner	<Name>	Approved (15-09-2026)